

Gerrymandering Metrics

Gerrymandering Analysis: 2024 vs AB 604

Setup

```
library(tidyverse)
library(scales)
```

Helper Functions

```
# Calculate wasted votes for two parties
calculate_wasted_votes <- function(votes_a, votes_b) {
  # votes_a and votes_b are vectors of votes by district
  total_votes <- votes_a + votes_b
  win_threshold <- total_votes / 2

  # For party A:
  # - In districts A wins: wasted = votes above threshold (surplus)
  # - In districts A loses: wasted = all votes
  wasted_a <- ifelse(votes_a > votes_b,
                     votes_a - win_threshold, # surplus in won districts
                     votes_a)                 # all votes in lost districts

  # For party B: same logic
  wasted_b <- ifelse(votes_b > votes_a,
                     votes_b - win_threshold,
                     votes_b)

  return(list(wasted_a = wasted_a, wasted_b = wasted_b))
}

# Calculate mean-median score
calculate_mean_median <- function(vote_shares) {
  # vote_shares: vector of vote shares for one party across districts
  mean_share <- mean(vote_shares, na.rm = TRUE)
  median_share <- median(vote_shares, na.rm = TRUE)
  return(mean_share - median_share)
}

# Calculate efficiency gap
calculate_efficiency_gap <- function(votes_a, votes_b) {
  wasted <- calculate_wasted_votes(votes_a, votes_b)
  total_votes <- sum(votes_a + votes_b)
```

```
gap <- (sum(wasted$wasted_a) - sum(wasted$wasted_b)) / total_votes
return(gap)
}

cat("✓ Helper functions loaded\n")
```

✓ Helper functions loaded

Part 4: 2024 Election Results and the 2024 District Map

Load and Prepare Data

```
# Load cleaned precinct data
precinct_data <- read_csv("data/g24_sov_by_g24_svprec_clean.csv",
show_col_types = FALSE)

# Load candidate information
candidates <- read_csv("data/g24-candidates-by-district.csv", show_col_types =
FALSE)

# Get congressional candidates only
cong_candidates <- candidates %>%
  filter(DISTRICT_TYPE == "CNG")

cat("Data loaded:\n")
```

Data loaded:

```
cat(" Precincts:", nrow(precinct_data), "\n")
```

Precincts: 51123

```
cat(" Congressional candidates:", nrow(cong_candidates), "\n\n")
```

Congressional candidates: 104

Map Candidates to Parties

```
# Create mapping of vote columns to parties
party_mapping <- cong_candidates %>%
  select(FIELD, DISTRICT, NAME) %>%
```

```
mutate(party = case_when(
  grepl("DEM", FIELD) ~ "Democrat",
  grepl("REP", FIELD) ~ "Republican",
  grepl("IND", FIELD) ~ "Independent",
  grepl("LIB", FIELD) ~ "Libertarian",
  grepl("GRN", FIELD) ~ "Green",
  TRUE ~ "Other"
))

cat("Party breakdown:\n")
```

Party breakdown:

```
print(table(party_mapping$party))
```

Democrat	Independent	Republican
54	1	49

```
cat("\n")
```

Aggregate Votes by Party and District (2024)

```
# Get congressional vote columns
cong_vote_cols <- names(precinct_data)[grep("^CNG", names(precinct_data))]

# Aggregate to district level
district_votes_2024 <- precinct_data %>%
  filter(!is.na(CDDIST)) %>%
  group_by(CDDIST) %>%
  summarise(across(all_of(cong_vote_cols), ~sum(., na.rm = TRUE)),
    .groups = "drop")

# Pivot to long format and join with party info
district_party_votes_2024 <- district_votes_2024 %>%
  pivot_longer(cols = all_of(cong_vote_cols),
    names_to = "FIELD",
    values_to = "votes") %>%
  left_join(party_mapping, by = "FIELD") %>%
  filter(votes > 0)
```

Warning in left_join(., party_mapping, by = "FIELD"): Detected an unexpected many-to-many relationship between `x` and `y`.

- i Row 1 of `x` matches multiple rows in `y`.
- i Row 1 of `y` matches multiple rows in `x`.
- i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

```
# Sum votes by district and party
party_totals_2024 <- district_party_votes_2024 %>%
  group_by(CDDIST, party) %>%
  summarise(total_votes = sum(votes), .groups = "drop") %>%
  pivot_wider(names_from = party, values_from = total_votes, values_fill = 0)

cat("2024 district-level results by party:\n")
```

2024 district-level results by party:

```
print(head(party_totals_2024, 10))
```

```
# A tibble: 10 × 4
  CDDIST Democrat Independent Republican
  <dbl>      <dbl>      <dbl>      <dbl>
1     0 2700597426      266700 1645017012
2     1   5634072         0    9991200
3     2  13891584         0    5107536
4     3   9585960         0   11226960
5     4  11593371         0    5502912
6     5   6857817         0   10282704
7     6   8434686         0    5838000
8     7  10065411         0    4717104
9     8  10289556         0    3404736
10    9   6634743         0    5808288
```

```
cat("\n")
```

Calculate Democrat vs Republican Metrics (2024)

```
# For gerrymandering metrics, focus on D vs R (exclude third parties for now)
dem_rep_2024 <- party_totals_2024 %>%
  select(CDDIST, Democrat, Republican) %>%
  mutate(
    total = Democrat + Republican,
    dem_share = Democrat / total,
    rep_share = Republican / total,
    winner = if_else(Democrat > Republican, "D", "R"),
```

```

    margin = abs(dem_share - rep_share)
  )
cat("=== 2024 Election Results ===\n\n")

```

```

=== 2024 Election Results ===

```

```

# Seat counts
seats_2024 <- table(dem_rep_2024$winner)
cat("Seat Distribution:\n")

```

```

Seat Distribution:

```

```

cat("  Democrats:", seats_2024["D"], "seats\n")

```

```

Democrats: 44 seats

```

```

cat("  Republicans:", seats_2024["R"], "seats\n\n")

```

```

Republicans: 9 seats

```

```

# Mean-median score
mm_score_2024 <- calculate_mean_median(dem_rep_2024$dem_share)
cat("Mean-Median Score:", round(mm_score_2024, 4), "\n")

```

```

Mean-Median Score: -0.0132

```

```

cat("  (Positive = Democrats packed/cracked, Negative = Republicans packed/
cracked)\n\n")

```

```

(Positive = Democrats packed/cracked, Negative = Republicans packed/cracked)

```

```

# Efficiency gap
eg_2024 <- calculate_efficiency_gap(dem_rep_2024$Democrat,
dem_rep_2024$Republican)
cat("Efficiency Gap:", round(eg_2024, 4), "\n")

```

Efficiency Gap: -0.2306

```
cat(" (Positive = Democrats waste more, Negative = Republicans waste  
more)\n\n")
```

(Positive = Democrats waste more, Negative = Republicans waste more)

```
# Wasted votes breakdown  
wasted_2024 <- calculate_wasted_votes(dem_rep_2024$Democrat,  
dem_rep_2024$Republican)  
total_wasted_dem <- sum(wasted_2024$wasted_a)  
total_wasted_rep <- sum(wasted_2024$wasted_b)  
  
cat("Wasted Votes:\n")
```

Wasted Votes:

```
cat(" Democrats:", comma(total_wasted_dem), "\n")
```

Democrats: 682,777,398

```
cat(" Republicans:", comma(total_wasted_rep), "\n")
```

Republicans: 1,851,759,591

```
cat(" Difference:", comma(total_wasted_dem - total_wasted_rep), "\n\n")
```

Difference: -1,168,982,193

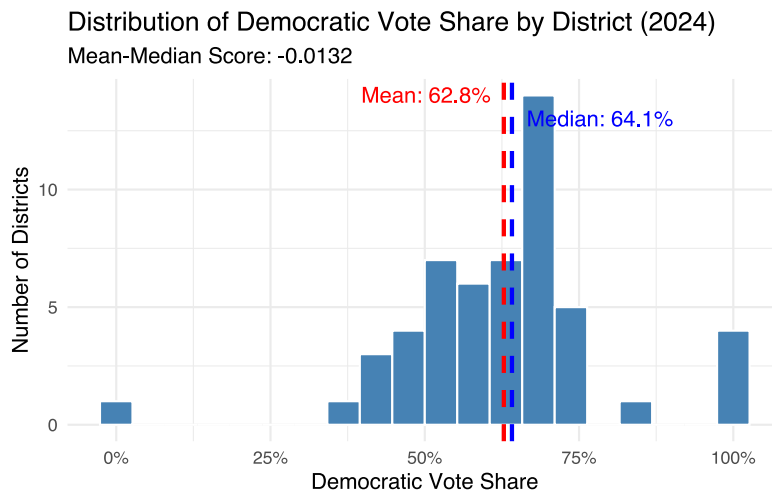
Visualize 2024 Results

```
# Vote share distribution  
ggplot(dem_rep_2024, aes(x = dem_share)) +  
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +  
  geom_vline(xintercept = mean(dem_rep_2024$dem_share),  
            color = "red", linetype = "dashed", linewidth = 1) +  
  geom_vline(xintercept = median(dem_rep_2024$dem_share),  
            color = "blue", linetype = "dashed", linewidth = 1) +  
  annotate("text", x = mean(dem_rep_2024$dem_share), y = Inf,
```

```

    label = paste0("Mean: ", percent(mean(dem_rep_2024$dem_share),
0.1)),
    vjust = 1.5, hjust = 1.1, color = "red") +
  annotate("text", x = median(dem_rep_2024$dem_share), y = Inf,
    label = paste0("Median: ", percent(median(dem_rep_2024$dem_share),
0.1)),
    vjust = 3, hjust = -0.1, color = "blue") +
  scale_x_continuous(labels = percent) +
  labs(title = "Distribution of Democratic Vote Share by District (2024)",
    subtitle = paste0("Mean-Median Score: ", round(mm_score_2024, 4)),
    x = "Democratic Vote Share",
    y = "Number of Districts") +
  theme_minimal()

```



Part 6: 2024 Election Results and the AB 604 District Map

Load AB 604 Results

```

# Load AB 604 district results (created in Part 5)
ab604_results <- read_csv("data/ab604_district_results.csv", show_col_types =
FALSE)

cat("AB 604 results loaded:", nrow(ab604_results), "districts\n\n")

```

AB 604 results loaded: 52 districts

Aggregate AB 604 Votes by Party

```

# Same vote columns, but for AB 604 districts
ab604_party_votes <- ab604_results %>%

```

```

pivot_longer(cols = all_of(cong_vote_cols),
              names_to = "FIELD",
              values_to = "votes") %>%
left_join(party_mapping, by = "FIELD") %>%
filter(votes > 0)

```

Warning in left_join(., party_mapping, by = "FIELD"): Detected an unexpected many-to-many relationship between `x` and `y`.
 i Row 1 of `x` matches multiple rows in `y`.
 i Row 1 of `y` matches multiple rows in `x`.
 i If a many-to-many relationship is expected, set `relationship = "many-to-many"` to silence this warning.

```

# Sum by district and party
party_totals_ab604 <- ab604_party_votes %>%
  group_by(AB604_DISTRICT, party) %>%
  summarise(total_votes = sum(votes), .groups = "drop") %>%
  pivot_wider(names_from = party, values_from = total_votes, values_fill = 0)

cat("AB 604 district-level results by party:\n")

```

AB 604 district-level results by party:

```

print(head(party_totals_ab604, 10))

```

```

# A tibble: 10 × 4
  AB604_DISTRICT Democrat Republican Independent
  <chr>          <dbl>      <dbl>      <dbl>
1 01             9309315.    7226278.         0
2 02            11830457.    7123247.         0
3 03             9686669.    7902613.         0
4 04             9656740.    6897002.         0
5 05             6023497.    9953875.         0
6 06             8721411.    7291402.         0
7 07             9655806.    6943818.         0
8 08            10513032.    4589420.         0
9 09             7672960.    5145961.         0
10 10            12315949.    5869883.         0

```

```

cat("\n")

```

Calculate Democrat vs Republican Metrics (AB 604)


```
# Focus on D vs R
dem_rep_ab604 <- party_totals_ab604 %>%
  select(AB604_DISTRICT, Democrat, Republican) %>%
  mutate(
    total = Democrat + Republican,
    dem_share = Democrat / total,
    rep_share = Republican / total,
    winner = if_else(Democrat > Republican, "D", "R"),
    margin = abs(dem_share - rep_share)
  )

cat("=== AB 604 Projected Results ===\n\n")
```

```
=== AB 604 Projected Results ===
```

```
# Seat counts
seats_ab604 <- table(dem_rep_ab604$winner)
cat("Seat Distribution:\n")
```

```
Seat Distribution:
```

```
cat("  Democrats:", seats_ab604["D"], "seats\n")
```

```
Democrats: 48 seats
```

```
cat("  Republicans:", seats_ab604["R"], "seats\n\n")
```

```
Republicans: 4 seats
```

```
# Mean-median score
mm_score_ab604 <- calculate_mean_median(dem_rep_ab604$dem_share)
cat("Mean-Median Score:", round(mm_score_ab604, 4), "\n")
```

```
Mean-Median Score: 0.0287
```

```
cat("  (Positive = Democrats packed/cracked, Negative = Republicans packed/
cracked)\n\n")
```

(Positive = Democrats packed/cracked, Negative = Republicans packed/cracked)

```
# Efficiency gap
eg_ab604 <- calculate_efficiency_gap(dem_rep_ab604$Democrat,
dem_rep_ab604$Republican)
cat("Efficiency Gap:", round(eg_ab604, 4), "\n")
```

Efficiency Gap: -0.1843

```
cat(" (Positive = Democrats waste more, Negative = Republicans waste
more)\n\n")
```

(Positive = Democrats waste more, Negative = Republicans waste more)

```
# Wasted votes
wasted_ab604 <- calculate_wasted_votes(dem_rep_ab604$Democrat,
dem_rep_ab604$Republican)
total_wasted_dem_ab604 <- sum(wasted_ab604$wasted_a)
total_wasted_rep_ab604 <- sum(wasted_ab604$wasted_b)

cat("Wasted Votes:\n")
```

Wasted Votes:

```
cat(" Democrats:", comma(total_wasted_dem_ab604), "\n")
```

Democrats: 114,694,513

```
cat(" Republicans:", comma(total_wasted_rep_ab604), "\n")
```

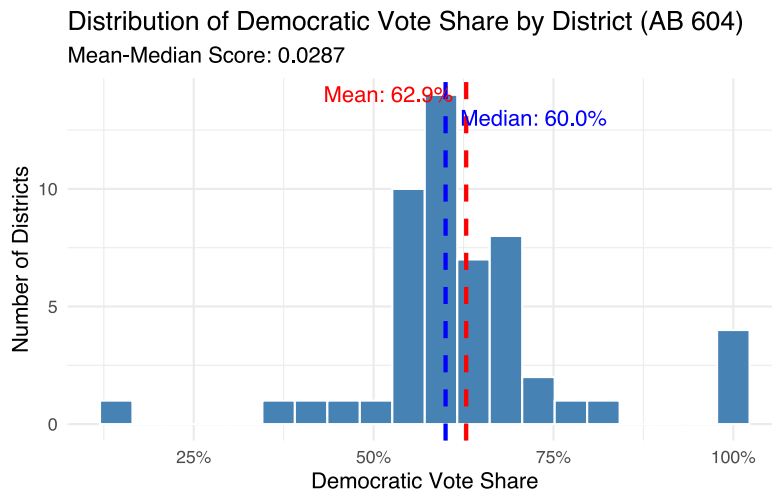
Republicans: 248,559,871

```
cat(" Difference:", comma(total_wasted_dem_ab604 - total_wasted_rep_ab604),
"\n\n")
```

Difference: -133,865,359

Visualize AB 604 Results

```
# Vote share distribution
ggplot(dem_rep_ab604, aes(x = dem_share)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  geom_vline(xintercept = mean(dem_rep_ab604$dem_share),
             color = "red", linetype = "dashed", linewidth = 1) +
  geom_vline(xintercept = median(dem_rep_ab604$dem_share),
             color = "blue", linetype = "dashed", linewidth = 1) +
  annotate("text", x = mean(dem_rep_ab604$dem_share), y = Inf,
          label = paste0("Mean: ", percent(mean(dem_rep_ab604$dem_share),
0.1))),
          vjust = 1.5, hjust = 1.1, color = "red") +
  annotate("text", x = median(dem_rep_ab604$dem_share), y = Inf,
          label = paste0("Median: ", percent(median(dem_rep_ab604$dem_share),
0.1))),
          vjust = 3, hjust = -0.1, color = "blue") +
  scale_x_continuous(labels = percent) +
  labs(title = "Distribution of Democratic Vote Share by District (AB 604)",
       subtitle = paste0("Mean-Median Score: ", round(mm_score_ab604, 4)),
       x = "Democratic Vote Share",
       y = "Number of Districts") +
  theme_minimal()
```



Comparison: 2024 vs AB 604

```
cat("\n===== \n")
```

```
=====
```

```
cat("COMPARISON: 2024 vs AB 604\n")
```

COMPARISON: 2024 vs AB 604

```
cat("=====\n\n")
```

```
=====
```

```
# Seat changes  
cat("Seat Distribution:\n")
```

Seat Distribution:

```
cat("                2024    AB 604    Change\n")
```

	2024	AB 604	Change
--	------	--------	--------

```
cat(" Democrats:      ", sprintf("%4d", seats_2024["D"]),  
    "      ", sprintf("%4d", seats_ab604["D"]),  
    "      ", sprintf("%+3d", seats_ab604["D"] - seats_2024["D"]), "\n")
```

Democrats:	44	48	+4
------------	----	----	----

```
cat(" Republicans:    ", sprintf("%4d", seats_2024["R"]),  
    "      ", sprintf("%4d", seats_ab604["R"]),  
    "      ", sprintf("%+3d", seats_ab604["R"] - seats_2024["R"]), "\n\n")
```

Republicans:	9	4	-5
--------------	---	---	----

```
# Metric changes  
cat("Gerrymandering Metrics:\n")
```

Gerrymandering Metrics:

```
cat("                2024    AB 604    Change\n")
```

2024	AB 604	Change
------	--------	--------

```
cat(" Mean-Median Score: ", sprintf("%.4f", mm_score_2024),
    " ", sprintf("%.4f", mm_score_ab604),
    " ", sprintf("%.4f", mm_score_ab604 - mm_score_2024), "\n")
```

Mean-Median Score:	-0.0132	0.0287	+0.0418
--------------------	---------	--------	---------

```
cat(" Efficiency Gap: ", sprintf("%.4f", eg_2024),
    " ", sprintf("%.4f", eg_ab604),
    " ", sprintf("%.4f", eg_ab604 - eg_2024), "\n\n")
```

Efficiency Gap:	-0.2306	-0.1843	+0.0464
-----------------	---------	---------	---------

```
# Interpretation
cat("Interpretation:\n")
```

Interpretation:

```
if (abs(mm_score_ab604) > abs(mm_score_2024)) {
  cat(" ⚠ Mean-median score INCREASED in magnitude\n")
  cat(" Suggests AB 604 may introduce more partisan advantage\n")
} else {
  cat(" ✓ Mean-median score DECREASED in magnitude\n")
  cat(" Suggests AB 604 may reduce partisan advantage\n")
}
```

⚠ Mean-median score INCREASED in magnitude
Suggests AB 604 may introduce more partisan advantage

```
if (abs(eg_ab604) > abs(eg_2024)) {
  cat(" ⚠ Efficiency gap INCREASED in magnitude\n")
  cat(" Suggests AB 604 may waste more votes unevenly\n")
} else {
  cat(" ✓ Efficiency gap DECREASED in magnitude\n")
  cat(" Suggests AB 604 may distribute votes more efficiently\n")
}
```

- ✓ Efficiency gap DECREASED in magnitude
Suggests AB 604 may distribute votes more efficiently

```
cat("\n")
```

Side-by-Side Comparison

```
# Combine data for comparison
comparison_data <- bind_rows(
  dem_rep_2024 %>% mutate(map = "2024", district = as.character(CDDIST)) %>%
  select(-CDDIST),
  dem_rep_ab604 %>% mutate(map = "AB 604", district =
  as.character(AB604_DISTRICT)) %>% select(-AB604_DISTRICT)
)

# Plot both distributions
ggplot(comparison_data, aes(x = dem_share, fill = map)) +
  geom_histogram(bins = 20, alpha = 0.6, position = "identity", color =
  "white") +
  geom_vline(data = data.frame(
    map = c("2024", "AB 604"),
    mean_val = c(mean(dem_rep_2024$dem_share), mean(dem_rep_ab604$dem_share))
  ), aes(xintercept = mean_val, color = map), linetype = "dashed", linewidth =
  1) +
  scale_fill_manual(values = c("2024" = "gray50", "AB 604" = "steelblue")) +
  scale_color_manual(values = c("2024" = "gray20", "AB 604" = "blue")) +
  scale_x_continuous(labels = percent) +
  labs(title = "Democratic Vote Share Distribution: 2024 vs AB 604",
    subtitle = "Dashed lines show mean values",
    x = "Democratic Vote Share",
    y = "Number of Districts",
    fill = "District Map",
    color = "Mean") +
  theme_minimal()
```

Democratic Vote Share Distribution: 2024 vs AB 604

Dashed lines show mean values

